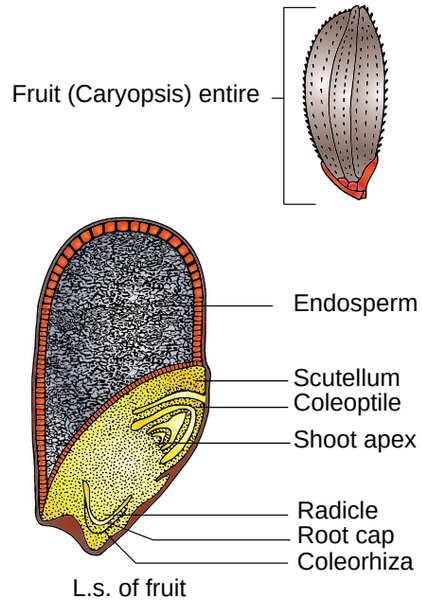


Figure 1.23(b) Monocot seed - *Oryza sativa*

Activity

Soak seeds of green gram for three hours. Drain the water and place few seeds in a clean tray containing moist cotton or filter paper. Allow the seeds to sprout. Collect the sprouted seeds, cut open and observe the parts. Record your observation.

1.7 Apomixis

Reproduction involving fertilization in flowering plants is called amphimixis and wherever reproduction does not involve union of male and female gametes is called apomixis.

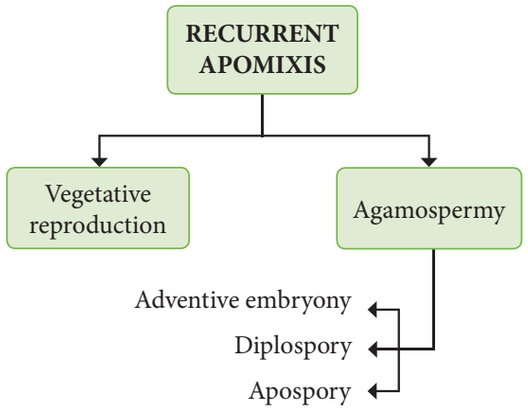
The term Apomixis was introduced by Winkler in the year 1908. It is defined as the substitution of the usual sexual system (Amphimixis) by a form of reproduction which does not involve meiosis and syngamy.

Maheswari (1950) classified Apomixis into two types - Recurrent and Non recurrent

Recurrent apomixis: It includes vegetative reproduction and agamospermy

Non recurrent apomixis: Haploid embryo sac developed after meiosis, develops into a embryo without fertilization.

The outline classification of Recurrent apomixis is given below.



Vegetative reproduction: Plants propagate by any part other than seeds

Bulbils – *Fritillaria imperialis*; Bulbs – *Allium*; Runner – *Mentha arvensis*; Sucker - *Chrysanthemum*

Agamospermy: It refers to processes by which Embryos are formed by eliminating meiosis and syngamy.

Adventive embryony

An Embryo arises directly from the diploid sporophytic cells either from nucellus or integument. It is also called **sporophytic budding** because gametophytic phase is completely absent. Adventive embryos are found in *Citrus* and *Mangifera*

Diplospory (Generative apospory): A diploid embryo sac is formed from megaspore mother cell without a regular meiotic division Examples. *Eupatorium* and *Aerva.*

Apospory: Megaspore mother cell (MMC) undergoes the normal meiosis and four megaspores formed gradually disappear. A nucellar cell becomes activated and develops into a diploid embryo sac. This type of apospory is also called somatic apospory. Examples *Hieracium* and *Parthenium.*

1.8 Polyembryony

Occurrence of more than one embryo in a seed is called polyembryony (Figure 1.24). The first case of polyembryony was reported in certain oranges by Anton von Leeuwenhoek in the year 1719. Polyembryony is divided into four categories based on its origin.

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